

FFGFT (T0) and Helix–Light–Vortex (HLV) Compared

A Structural Side-by-Side: Substrate, Triadic Time, Spectral Fork

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Abstract

This work places FFGFT (T0, time–mass duality) and the Helix–Light–Vortex programme (HLV, M. Krüger) side by side. Both are *layered effective geometric programmes*: they replace the smooth continuum with a discrete, higher-dimensionally generated substrate, carry a three-fold (triadic) core motif, read particles as topological objects, root in classical proportion/harmonic geometry, and share a methodology of recovery limit plus null-model falsification. We separate *genuine isomorphisms* from *surface analogies*. The central difference: FFGFT carries *one* dimensionless constant $\xi \approx 1/7500$ with forward dimensionless predictions (fine-structure constant, Koide relation, lepton mass ratios), whereas HLV has no ξ -analogue but an explicitly reproducible aperiodic substrate together with a graph-Laplacian testbed against null models. The geometric fork *aperiodic quasicrystal* (HLV) versus *periodic T^4 torus with a fractal correction* (FFGFT) yields a measurable spectral difference, from which a *distinguishing* prediction is formulated. HLV's consciousness/biogravimetry layer is explicitly excluded from this physics comparison. As a shared guardrail we stress the triviality of any representation $X = \varphi^N$ or $X = \xi^N$: forward content lies solely in dimensionless ratios that are isolated, predicted, and precise; an absolute scale remains a calibrated input.

.1 Motivation and scope

HLV is now available in a deliberately *conservative*, falsifiable form in two recent documents: a “master explanation” (Zenodo 10.5281/zenodo.20596861) and a reproducible “6D-to-3D cut-and-project graph testbed” (Zenodo 10.5281/zenodo.20275888). The move made there – away from “spacetime is quasicrystalline” towards “can this substrate be made explicit, reproducible, and distinguishable from simpler null models?” – is methodologically the same turn that FFGFT performed in its corrections register (Doc. 190, notably the corpus-wide cleanup of the H_0 over-claims).

The aim here is a *comparison*, not a merger: to test where the kinship is structurally load-bearing, where it is merely verbal, and at which point the two programmes must *diverge* empirically.

.2 Common axes (overview)

Axis	FFGFT / T0 (Pascher)	HLV (Krüger)
Substrate / projection	compact 4-torus T^4 + fractal correction $D_f = 3 - \xi$; reduction chain $T^4 \rightarrow T^0$	aperiodic quasicrystal via cut-and-project ($\mathbb{Z}^5 \rightarrow 3D$), golden-ratio/Fibonacci
Time structure	time–mass duality $\tilde{T} \cdot m = 1$; Z_3 in the triangle reduction ($\det = 1 - \xi$)	triadic spiral time $\psi(t) = t + i\varphi(t) + j\chi(t)$, $i^2 = j^2 = -1$
Particle ontology	winding quantum numbers on T^4 ; Bell topological, not ontologically nonlocal	vortices / standing-wave modes of a scalar field Ψ in the lattice
Generative core	<i>one</i> dimensionless $\xi \approx 1/7500$	no ξ -analogue; substrate geometry (φ , Fibonacci)
Forward predictions	α , Koide, lepton ratios (dimensionless, isolated)	deliberately withheld; focus “is the substrate distinguishable?”
Recovery limit	$\hbar, c, G$ as conversions; $O(\xi)$ lattice dispersion; bridges (with honest negatives)	must recover QFT/GR in the limit; graph Laplacian vs. null models
Falsification channel	mode spectrum (T^4): CMB peak <i>ratios</i> , $B = 1/3$ from $D = 4$	graph-Laplacian spectrum vs. null model; vortex benchmarks

.3 Genuine isomorphisms vs. surface analogies

1. **Projection from higher dimensions.** *Resonance:* both read observed 3-space as the projection of a structured, higher-dimensional, discrete geometry – HLV via cut-and-project, FFGFT via the explicit chain $T^4 \rightarrow T^0$ (Doc. 270). *But:* *aperiodic* (quasicrystal) versus *periodic* (T^4) are different spectral classes – see Section 6. The motif is isomorphic, the realisation is not.
2. **Triadic core motif.** *Resonance:* HLV's hypercomplex time $\psi = t + i\varphi + j\chi$ sits next to the $\mathbb{Z}_3$ structure of the FFGFT triangle reduction ($\det(A_\xi) = 1 - \xi$, Doc. 206) and the three operation types of the reduction chain. *But:* HLV's triad is a *time algebra*, FFGFT's a *reduction-matrix symmetry*. The shared "3" is striking but not yet a proven isomorphism – a candidate for closer scrutiny, not an identity.
3. **Particles as topological objects.** *Resonance:* HLV vortices / standing modes correspond to FFGFT winding numbers on the torus; both relocate "quantum number" into a topological winding of the substrate. In FFGFT this also renders the Bell correlation as topological (not ontologically nonlocal). Here the kinship is most likely genuine.
4. **Harmonic/proportion root.** *Resonance:* HLV anchors in the golden ratio and Fibonacci; FFGFT in the Euler Tonnetz and the comma structure of temperament. Same instinct ("proportion/harmony $\rightarrow$ fundamental geometry"), different constant.
5. **Methodology.** *Resonance (strongest):* recovery limit plus null-model falsification. HLV's "distinguishable from simpler null models" is exactly the criterion by which FFGFT tested whether, e.g., $63/4$ is *isolated* or merely one of several "nice fractions" in a dense candidate field (Doc. 190, R35).

.4 Divergences and complementarity

- **Generative core.** FFGFT has a single dimensionless seed ξ and from it *forward* predictions ($\alpha = \xi(E_0/\text{MeV})^2$, Koide, lepton ratios). HLV (in the documents presented) has no such analogue and deliberately stays at "is the substrate distinguishable". $\Rightarrow$ FFGFT brings the seed, HLV the reproducible substrate construction – complementary, not competing.
- **Tool.** HLV's graph-Laplacian testbed against null models is a numerical falsification machinery that FFGFT could adopt for its own T^4 mode spectrum.
- **Substrate.** Aperiodic vs. periodic + fractal – a real geometric fork (Section 6).
- **Foundational maturity.** HLV has drawn external-review criticism of an under-specified, non-closed algebra and undefined operators; FFGFT works with a three-layer discipline (core derivations / algebraically proven bridges / plausibility reductions) and the Hilbert-space bridge (Doc. 230–232).
- **Scope.** HLV includes a consciousness/biogravimetry layer. This is *explicitly excluded* here: the present comparison concerns only the physical-geometric layers.

.5 Distinguishing prediction: the spectral fork

The scientifically robust core of the comparison is spectral. A *periodic* substrate (T^4) produces a discrete spectrum with integer mode labels $|n|^2$ and a harmonic, ladder-like structure; FFGFT reads off dimensionless CMB peak ratios $1 : 2.449 : 3.742 : \dots$, fixed by $D = 4$ (curvature parameter $B = 1/3$). An *aperiodic* substrate (quasicrystal), by contrast,

typically produces a hierarchical, gap-rich spectrum (singular-continuous density of states, Cantor-like gap structure) *without* a simple integer harmonic ladder.

These two signatures are *distinguishable*. Concrete (scale-free) test:

1. Run HLV's testbed with (a) the quasicrystal *and* (b) a *periodic* lattice null model – not only randomised nulls. The sharp question is aperiodic-vs-periodic, not discrete-vs-continuum.
2. Report the leading *deviation exponent* in the recovery limit: does the graph-Laplacian dispersion approach the continuum $\omega \propto |k|$, and is the residual $O(\text{small})$ and sign-definite? (FFGFT does exactly this with $O(\xi)$ corrections.)
3. Compare scale-free *spectral ratios*: the harmonic T^4 ratio spectrum against the quasicrystal gap spectrum – and both against data (CMB peaks resp. lattice benchmarks). Whichever ratio spectrum matches observation discriminates the substrates.

.6 Shared methodological guardrail

For *both* programmes: every magnitude can be written as φ^N or ξ^N ; a single measured scale in an exponent is therefore *not* a prediction. Forward content lies solely in dimensionless ratios that are simultaneously *isolated*, *predicted*, and *precise*. The absolute scale (in FFGFT H_0/R_H via the fitted exponent 41/4) remains a calibrated input, not evidence (Doc. 190, R35). An honest HLV prediction should likewise be a *scale-free* ratio from the graph-Laplacian spectrum, not an absolute magnitude dressed in φ .

.7 Status and next steps

This work is a structural mapping, not a unification claim. It remains open whether the triadic resonance is a genuine isomorphism or a coincidence, and whether either substrate matches the observed mode spectrum better. The obvious next step is a *joint* testbed run: HLV's graph Laplacian against a periodic null lattice, plus the scale-free ratio comparison with the T^4 spectrum. Only then – falsification before interpretation – is a statement about the kinship of the two geometries defensible.

.8 Sources

- M. Krüger, *A Conservative Master Explanation of the Helix–Light–Vortex Framework*, Zenodo (2026), doi:10.5281/zenodo.20596861.
- M. Krüger, *A Reproducible 6D-to-3D Cut-and-Project Graph Testbed for Quasicrystalline Laplacian Recovery and Vortex Benchmarks*, Zenodo (2026), doi:10.5281/zenodo.20275888.
- FFGFT: Doc. 190 (corrections register, R35), Doc. 206 (triangle reduction, Z_3), Doc. 230–232 (Hilbert-space bridge), Doc. 268 (CMB peaks from T^4), Doc. 270 (reduction chain $T^4 \rightarrow T^0$).